AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 2 · STUDENT LESSON

A Distribution of Possibilities

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Validate the table. Multiply the path. Name the event.

Compare conditional token paths while preserving scope and factual evidence limits.

Keep one habit: Probability contract → six paths → greedy counterexample → sampling/exact audit → independent table.

MAIN LESSON · SECTION 1

Which path is more likely, and what does that mean?

> Validate distributions and multiply conditional probabilities without treating them as factual evidence.

Your learning goals

I can check whether a next-token table is a valid **probability distribution**.

I can calculate a two-step **sequence probability** using conditional probabilities.

I can explain why token probability does not measure factual truth.

Recall stated alternatives and conventions

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why must token counts name the vocabulary and special-token convention?

Your end goal

By the end, I can multiply conditional probabilities to compare paths instead of only first tokens.

▼ Representation extension · IDs and meaning

Do adjacent vocabulary IDs prove that their tokens have similar meanings?

> **My readiness explanation**

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 2: A **Distribution** of Possibilities

A generator assigns different **likelihood**s to possible **continuation**s.



CREATOR MISSION

Run a weighted-word simulation and compare repeated **output**s.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Embeddings and position

This week’s end goal

MAIN LESSON · SECTION 2

Keep the prefix and alternatives beside each probability

> Weights need an explicit normalisation rule before they are probabilities.

The complete Status distribution

A constructed model uses prefix “Status:” and vocabulary alternatives ready, busy and unknown. Its next-token table is ready 0.5, busy 0.3, unknown 0.2. All entries are nonnegative and sum to 1.

A probability belongs to a token under this prefix and model, not to that token in every situation. An omitted vocabulary alternative would make the interpretation incomplete unless its probability were explicitly zero.

Detect an invalid distribution and normalise only permitted weights

A table containing 0.6,0.3,0.3 sums to 1.2 and is not already a **probability distribution**. If the values are nonnegative weights and the rule says normalise, divide each by 1.2. Do not silently treat weights as probabilities.

Negative weights cannot be repaired by this simple normalisation rule.

▼ **Representation extension · Token lookup and position scheme**

Use a two-token vocabulary: ID 1 means red with **embedding** [1,0]; ID 2 means blue with **embedding** [0,1]. This lookup table is constructed, not trained.

The **input** red blue has **embedding** rows [1,0] and [0,1], so its shape is two sequence positions by two components. Reversing the text reverses the rows. ID 2 is not “twice as blue” as ID 1 is red.

A model needs a way to use order. A collection of vectors without any order-sensitive procedure may lose that distinction. We introduce one explicit toy scheme: add **position vector** [0,0] at position 0 and [2,0] at position 1.

These invented values teach arithmetic; they are not a recommended production encoding.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use a **distribution** of possibilities to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Embeddings and position

MAIN LESSON · SECTION 3

Multiply down each branch, then add the stated paths

> Conditional products use the appropriate expanded prefix; they do not assume independence.

Read the complete second-step Status table

Expanded prefix	End probability	Today probability
Status: ready	0.8	0.2
Status: busy	0.5	0.5
Status: unknown	1	0

Compute all six paths and their total

The probability of ready followed by end is 0.5*0.8 = 0.4. Ready followed by today is 0.5*0.2 = 0.1. Busy followed by end is 0.3*0.5 = 0.15, and busy followed by today is 0.15.

Unknown followed by end is 0.2 and unknown followed by today is 0. Together these six two-step paths sum to 1. These products are chain-rule calculations; they do not assume the second token is independent of the first.

Trace the three end paths

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Ready probability 0.5	End given ready 0.8	Joint path probability 0.4
Busy probability 0.3	End given busy 0.5	Joint path probability 0.15
Unknown probability 0.2	End given unknown 1	Joint path probability 0.2

The second probability uses the expanded prefix on that branch.

Explain the deciding step

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Representation extension · Ordered rows and shape contract

Sequence	Position 0 representation	Position 1 representation
red blue	[1,0] + [0,0] = [1,0]	[0,1] + [2,0] = [2,1]
blue red	[0,1] + [0,0] = [0,1]	[1,0] + [2,0] = [3,0]

The same red token now has [1,0] at position 0 and [3,0] at position 1. Both token identity and position affect the represented row.

Addition requires equal widths; adding a three-component **position vector** to a two-component **embedding** violates the **input** contract. Record the position numbering and **embedding**-table version.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
ID 1 at position 0	[1,0] + [0,0]	Representation [1,0]
ID 1 at position 1	[1,0] + [2,0]	Representation [3,0]
Two different sequences	Inspect ordered rows	Do not infer meaning from ID gaps

The position contribution changes a token representation according to its location.

▼ Earlier wording, alternatives and picture transcripts

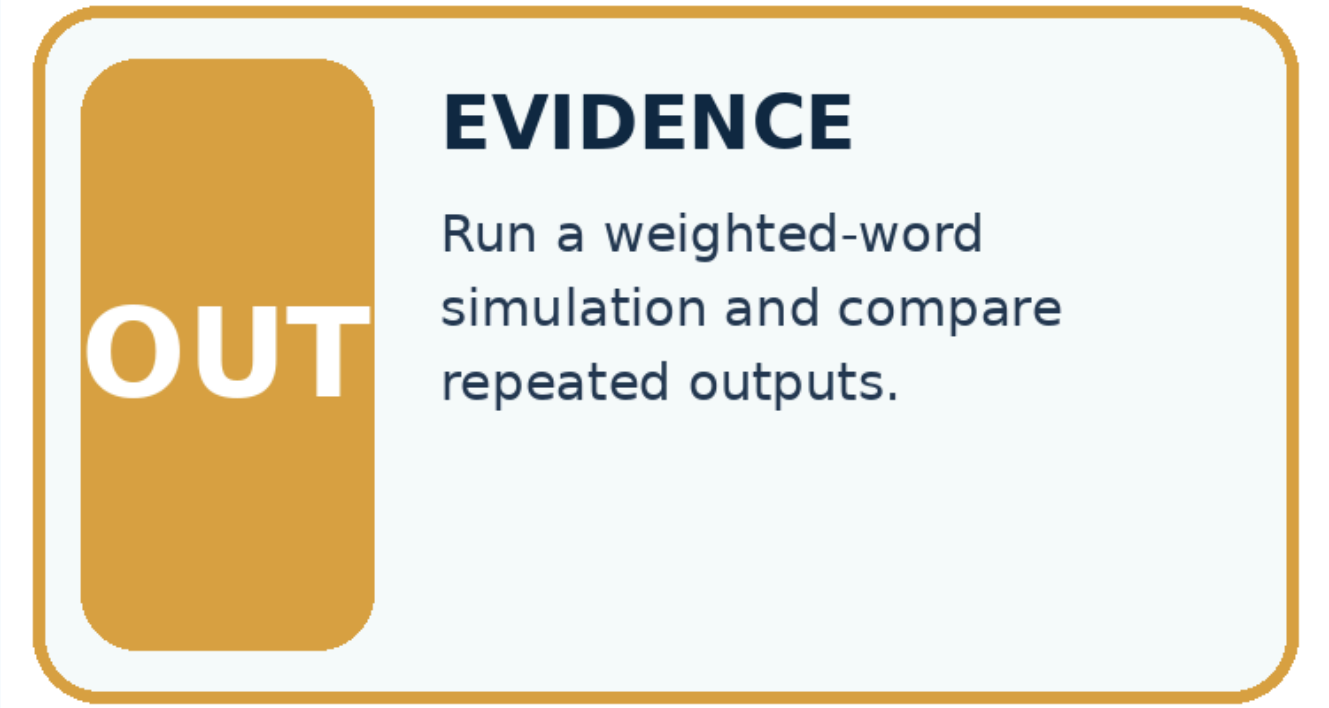
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Run a weighted-word simulation and compare repeated outputs.
↓
SYSTEM
A generator assigns different likelihoods to possible continuations.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

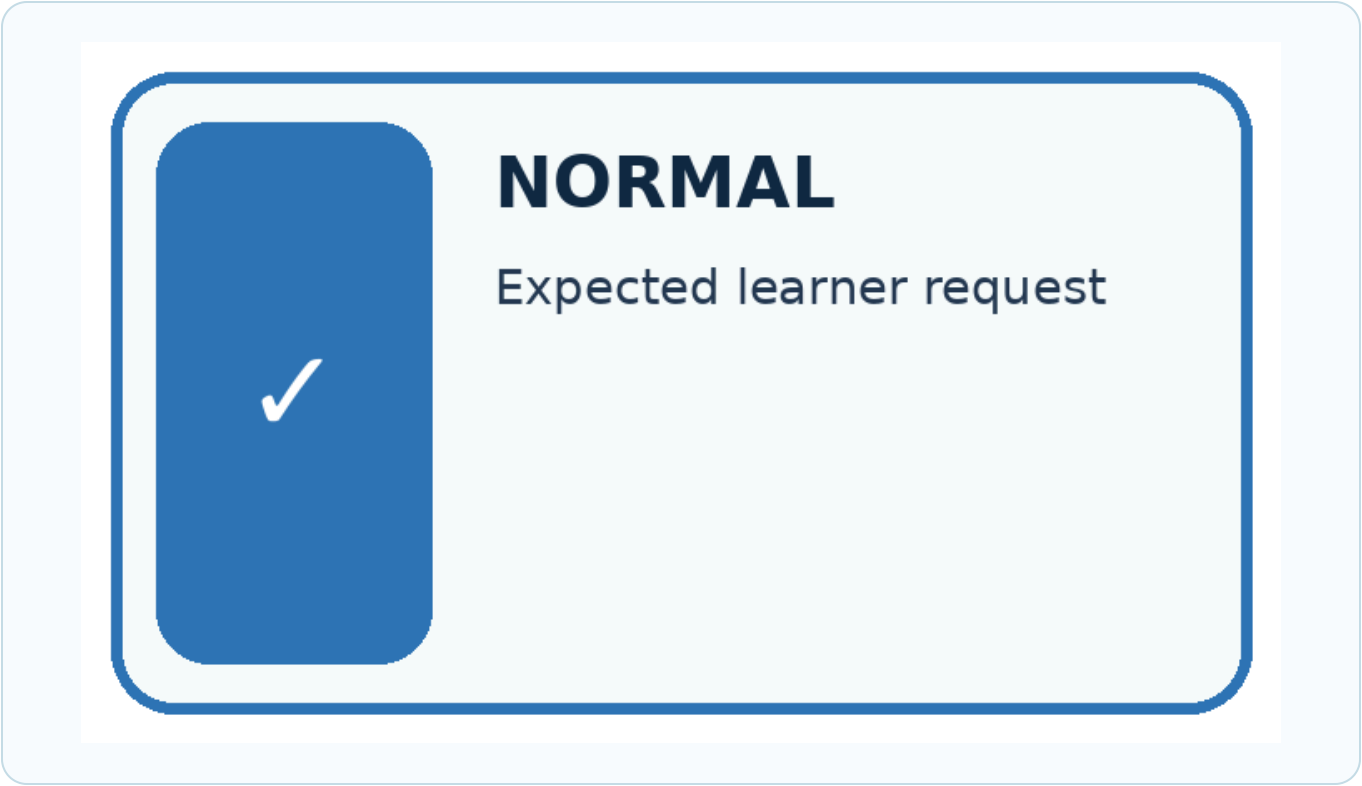
Apply the connected idea: Embeddings and position

Connect the picture and the explanation

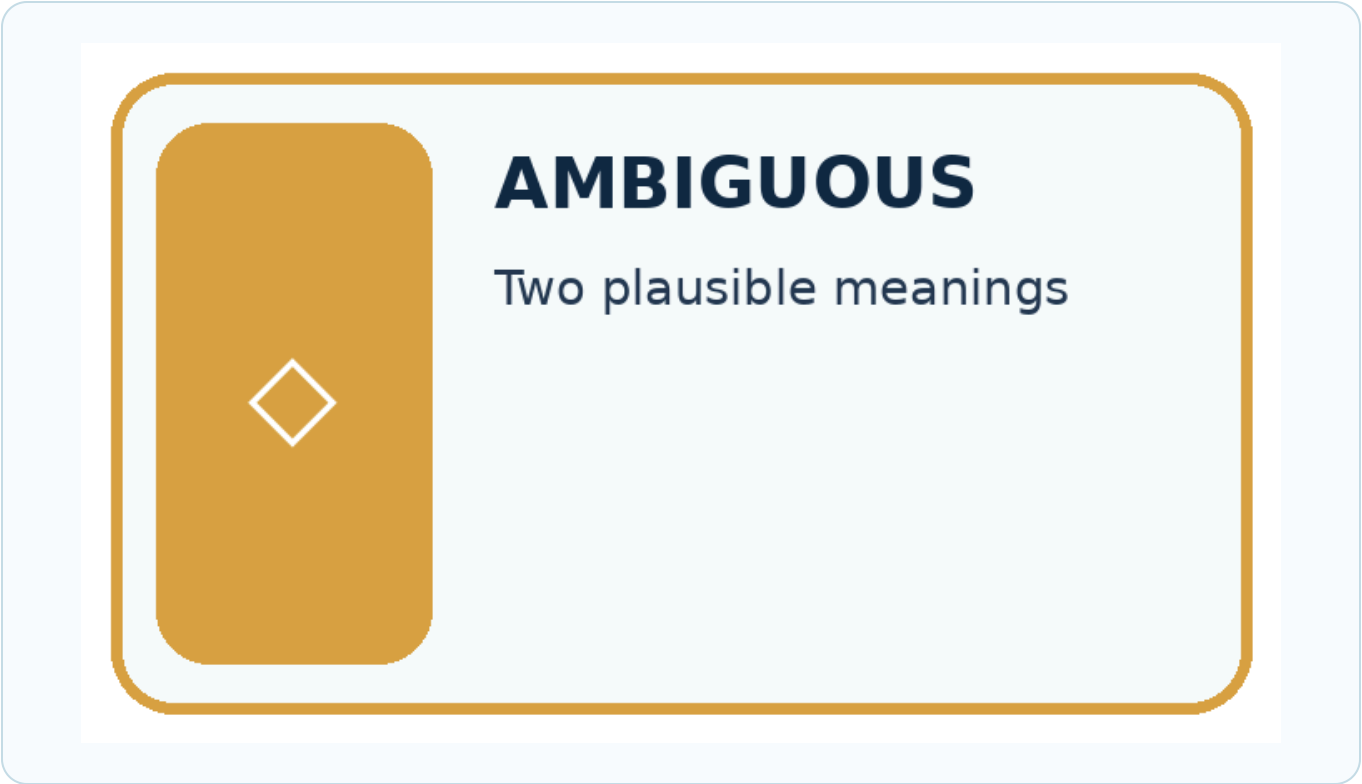
Compare valid tables, incomplete paths and unsupported claims

> A high-probability continuation is still an output to evaluate.

Four pictures with a purpose



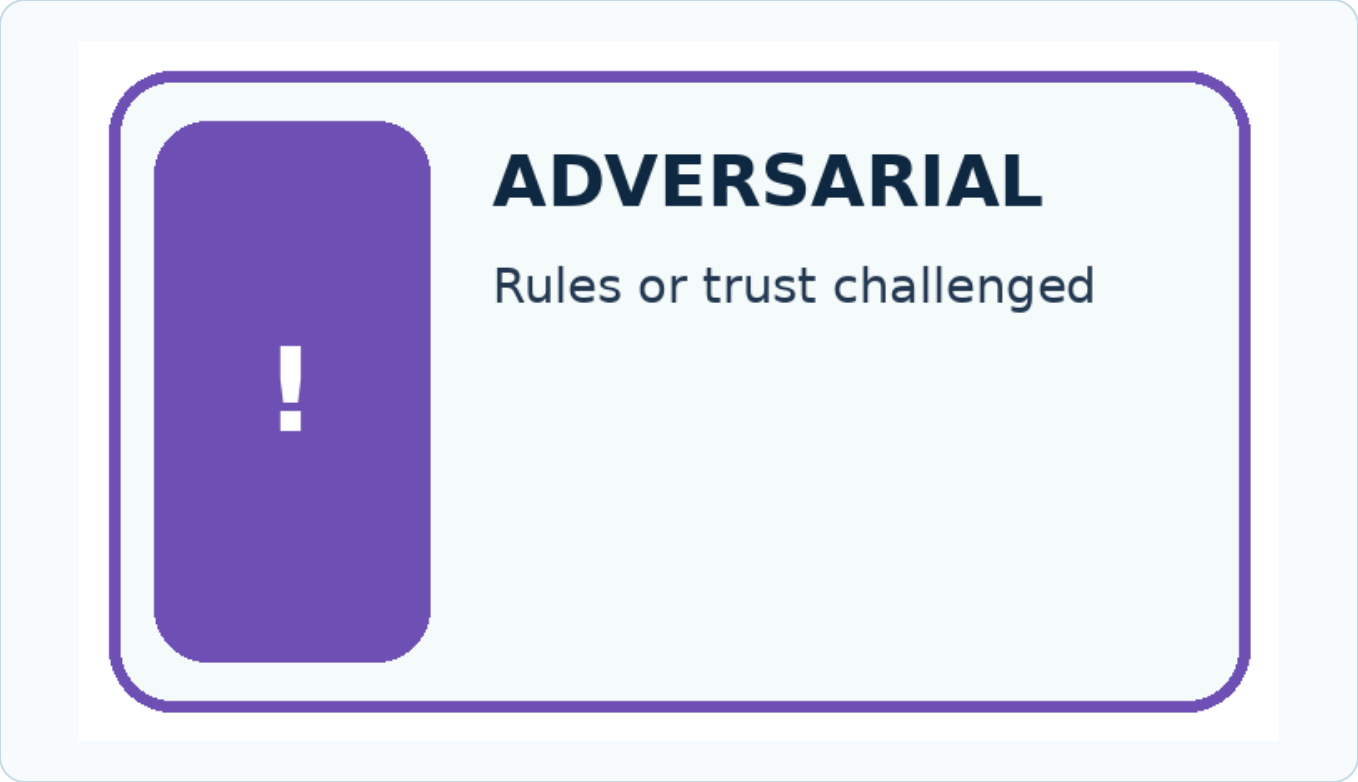
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

▼ Check after predicting

Prediction: sun→end becomes 0.44, beating unchanged rain→end 0.405.

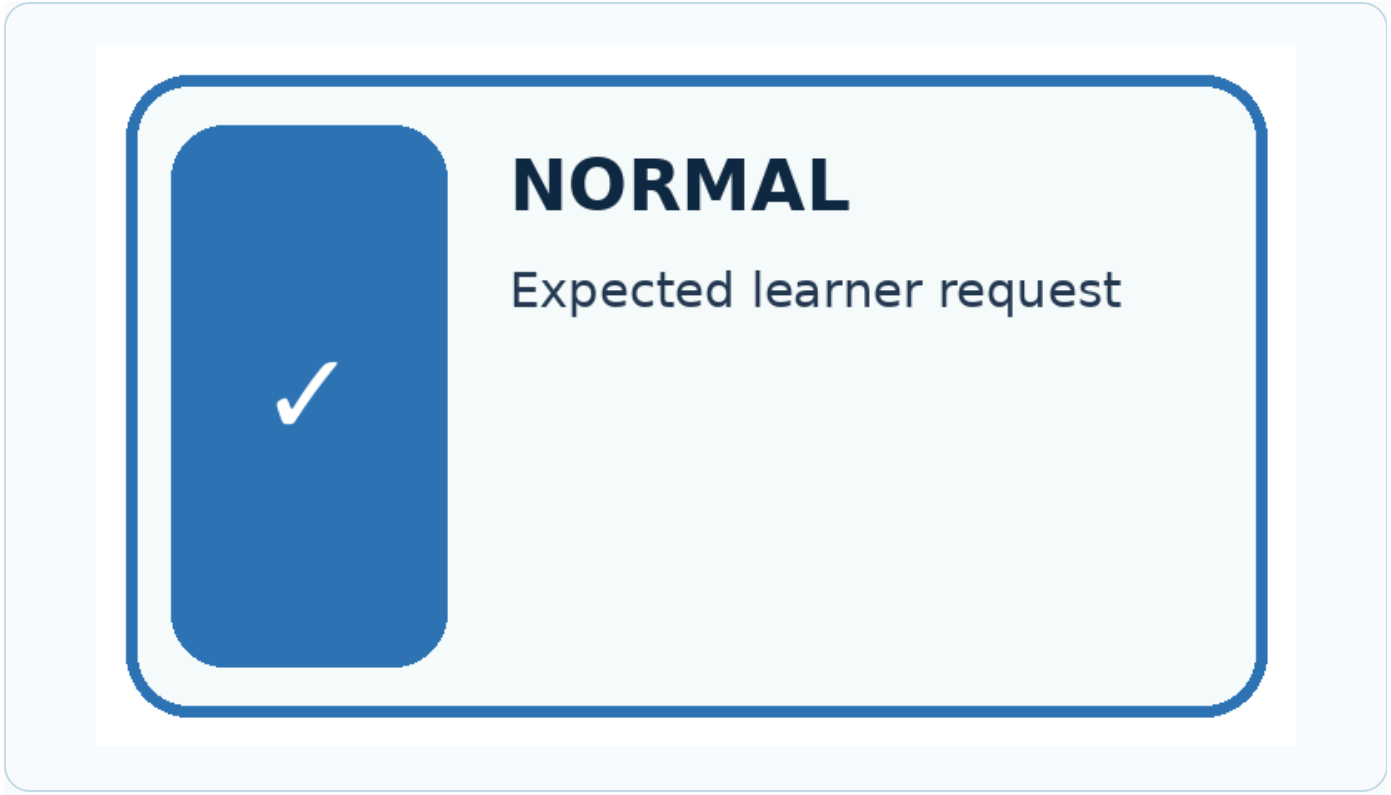
Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

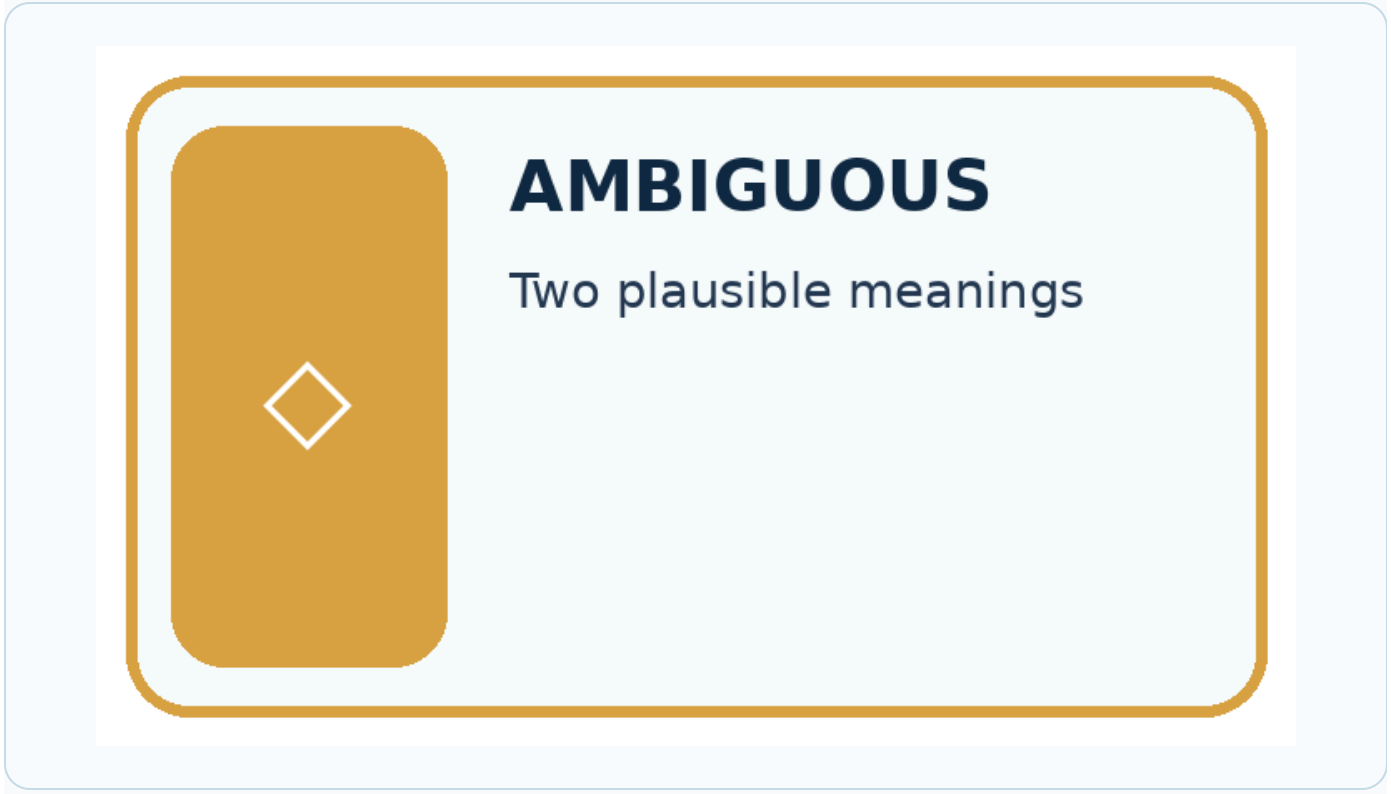
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Challenge greedy decoding, then trace a separate sun/rain tree

> Keep Status, A/B and sun/rain distributions separate.

Read the complete A/B counterexample

Greedy decoding chooses the largest next-token probability at each step, with an explicit tie rule if needed. It does not always maximise the probability of a complete sequence. Counterexample: first-step A=0.6 and B=0.4.

Given A, end=0.51 and x=0.49; given B, end=1. Greedy produces A,end with probability 0.306. The path B,end has probability 0.4, larger than 0.306.

This comparison concerns exactly two-step paths; longer completion probabilities require the later conditionals too.

Separate path likelihood from factual support

A high path probability measures the model's preference over these specified paths. It is not **evidence** that a status claim matches a live **observation**. Format checks and source checks remain separate.

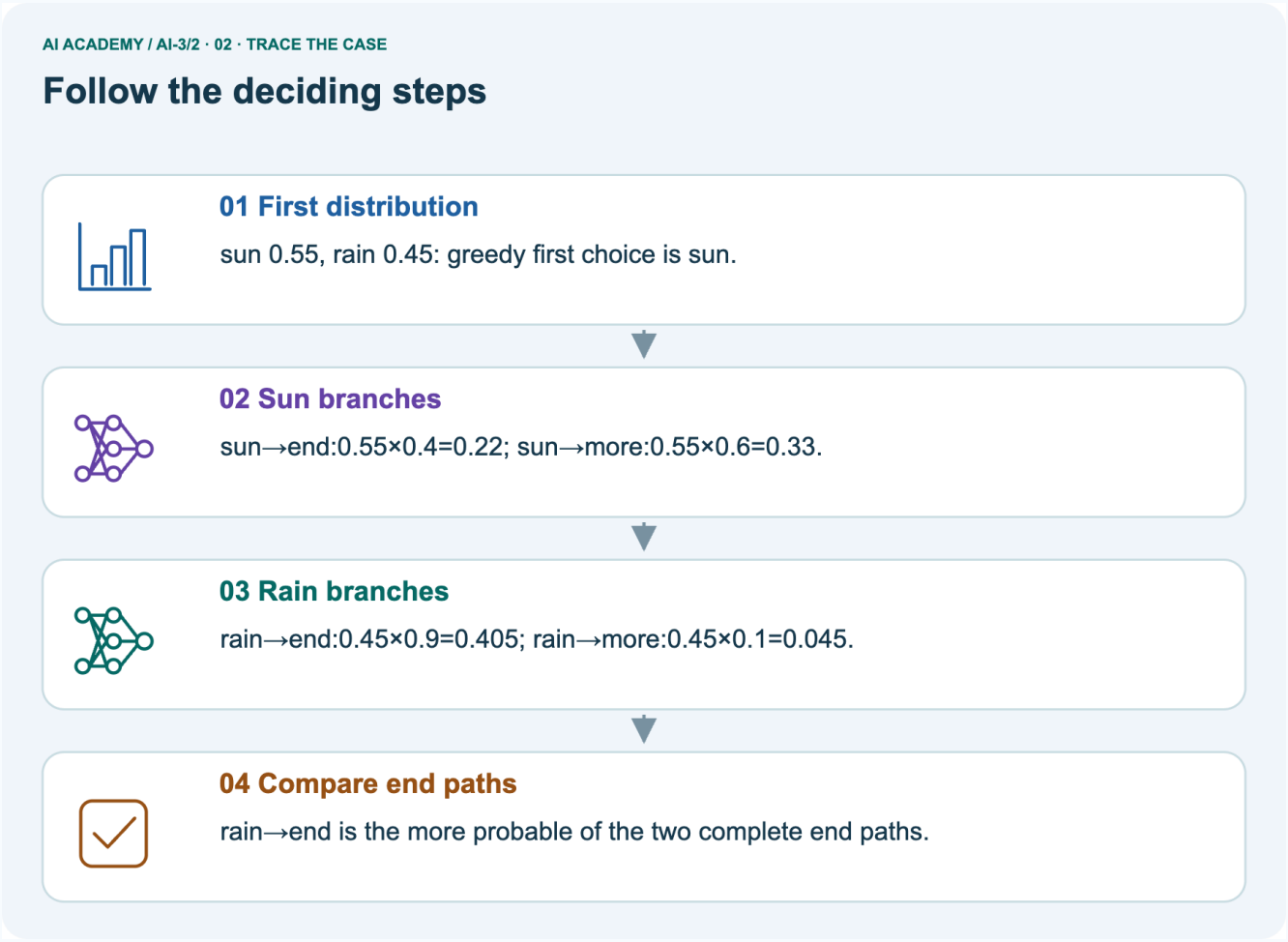
Read the separate pictured sun/rain branches

A fictional two-stage token tree illustrates next-token **prediction**.

First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

Trace all four branch products



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First distribution	sun 0.55, rain 0.45: greedy first choice is sun.
2. Sun branches	sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33.
3. Rain branches	rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045.
4. Compare end paths	rain→end is the more probable of the two complete end paths.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

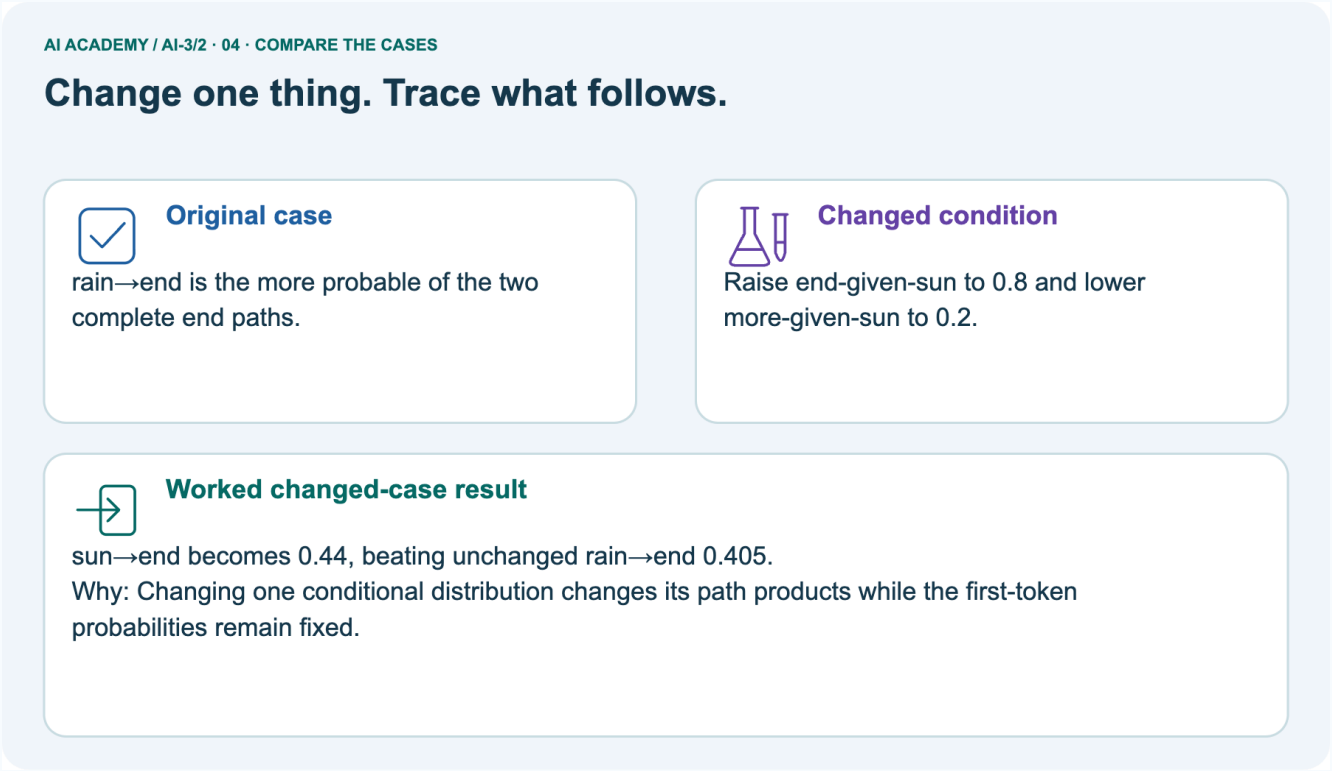
Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Predict changing only sun’s conditional distribution

- Change one thing: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the partial end-path sum

- These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.
- Read the labelled picture: Why is 0.625 not a broken probability sum?

▼ Representation extension · Row sum loses order

The elementwise sum of the represented rows is [3,1] for both red blue and blue red. This aggregate loses the row-order distinction despite the position additions.

A later computation must actually use the represented rows appropriately; merely adding **position vector**s does not guarantee that every summary preserves order.

A position scheme may use learned values or a deterministic construction. The original Transformer added positional information to **embedding**s. Modern architectures can use other order-sensitive schemes.

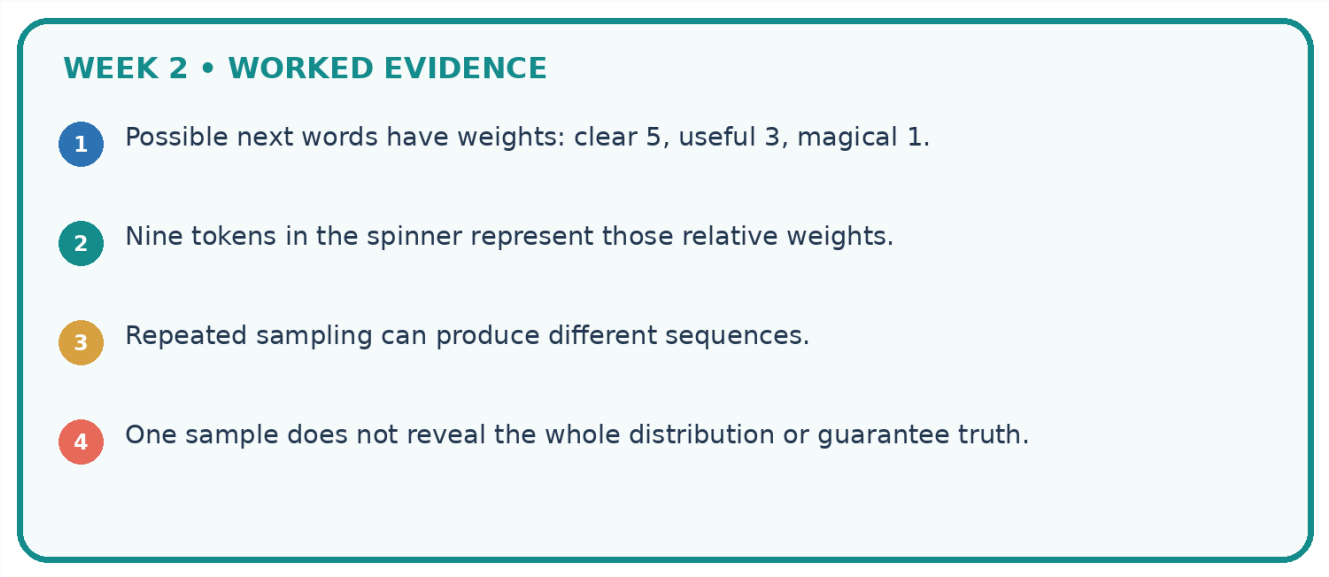
Our two-vector example demonstrates one possible **mechanism** and one information-loss failure, not all architecture choices.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Possible next words have weights: clear 5, useful 3, magical 1.

2. Nine tokens in the spinner represent those relative weights.

3. Repeated sampling can produce different sequences.

4. One **sample** does not reveal the whole **distribution** or guarantee truth.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

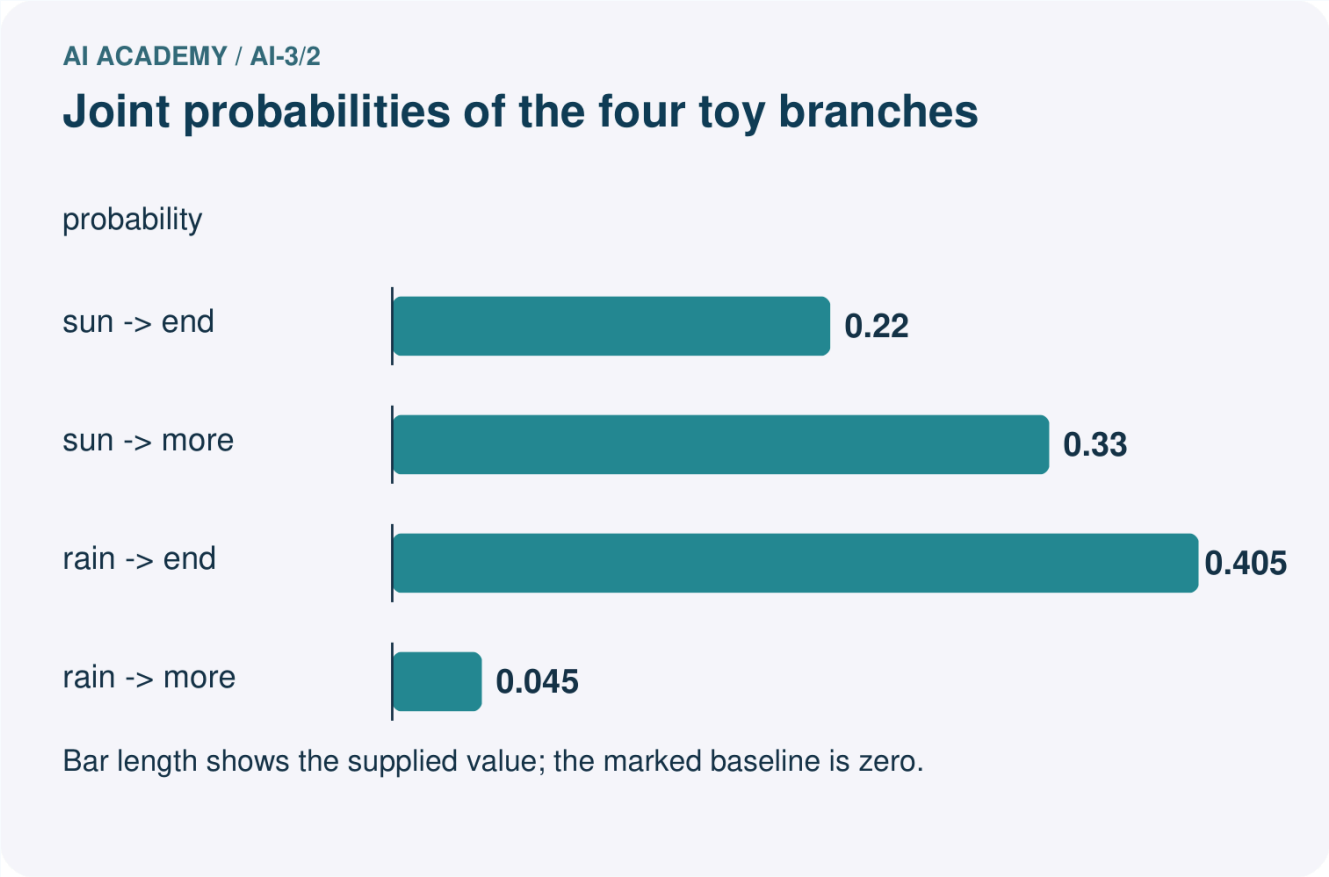
Apply the connected idea: Embeddings and position

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: A Distribution of Possibilities

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/2

A Distribution of Possibilities

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First distribution

sun 0.55, rain 0.45: greedy first choice is sun.



2. Sun branches

sun -> end:0.55 x 0.4=0.22; sun -> more:0.55 x 0.6=0.33.



3. Rain branches

rain -> end:0.45 x 0.9=0.405; rain -> more:0.45 x 0.1=0.045.



4. Compare end paths

rain -> end is the more probable of the two complete end paths.

CHANGE ONE THING

Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find which alternatives a probability sum includes

> A partial event need not have probability1.

Repair the incomplete-sum argument

Claim to repair: The two end paths sum to0.625, so the sun/rain table is invalid. That sum includes only sun→end and rain→end. The two more paths carry0.33+0.045=0.375, so all four two-step paths sum to1. The more paths have not ended; their eventual completed-sequence probabilities need later conditional tables that are not supplied.

Do not invent those probabilities or compare a partial branch with an unspecified full continuation.

Propose a revealing changed table

> **Exit: explain two errors in “ready has probability 0.5, so busy also has probability 0.5 and the room is ready half the time”.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: The model always chooses the single most likely continuation. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Always report the prefix, alternatives and stage. The complement of ready in the first table is busy or unknown, probability 0.5; it is not automatically busy.

Values rounded for display may not sum exactly to 1, so retain full values for calculations and label rounding. A missing or invalid table should lead to an explicit error, not a confident probability claim.

Apply the connected idea: Embeddings and position

State whether a vector is a token **embedding**, a **position vector** or their combination before calculating distances. Position changes can affect numerical similarity even for the same token.

An unknown token or a sequence beyond the toy position table needs explicit handling. This two-position contract rejects a third position; it does not silently wrap back to position 0.

> **Exit: a program compares vocabulary IDs instead of vectors and silently repeats its two-position table for a longer sequence. Identify both contract errors.**

MAIN LESSON · SECTION 8

Practise a separate red/blue path comparison

> State which prefix each conditional belongs to.

Read the supplied red/blue probabilities

Supported table: first red=0.4, blue=0.6. End given red=0.9 and end given blue=0.5. The red,end path has probability 0.36. The blue,end path has probability 0.3 despite blue being the larger first-step probability.

Compute both specified end paths

Guided: compute both products and distinguish the preferred first token from the more probable specified end path.

Complete main Status W2

Practise with less help: Calculate the paths

Use workbook W2 for the connected example “Next token **distribution**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair the independence claim

- Use the worked case for Next token **distribution**s. Proposed explanation: “Conditional products assume independence”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the pictured partial sum

- Why is 0.625 not a broken probability sum?

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Representation extension · Compute ordered rows and sums

Supported case: **embedding**s A=[2,1], B=[1,3], positions [0,0] and [1,-1]. Sequence A B becomes [2,1] and [2,2]. Sequence B A becomes [1,3] and [3,0]. Both sums are [4,3].

> **Guided: compute both ordered representations and compare their sums. Explain which operation discarded the difference.**

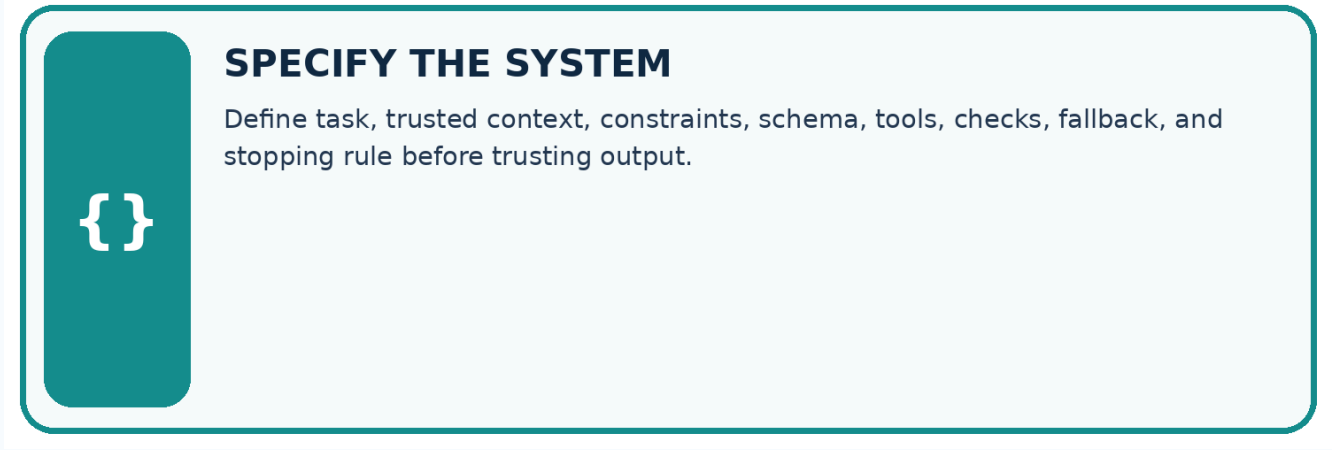
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A **position vector** guarantees every summary preserves order

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Embeddings and position

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Practise the connected topic: Embeddings and position

Run a specified sampling activity and an exact-path audit

> Keep sampled frequencies, expected counts and exact model probabilities distinct.

Nine-slip materials and repeatable procedure

1. Paper weighted-token simulation: make nine equal slips numbered1–9: clear on1–5, useful on6–8, magical on9. Mix thoroughly, draw one without looking, record its label, replace it and mix again. Repeat20times. The idealised probabilities are5/9,3/9,1/9; expected counts are20×those fractions, not required integer outcomes. Compare actual tallies with these expectations and record your sampling procedure.

   A twenty-draw difference does not prove the probabilities changed or that a token is true.

Predict the conditional-path outputs

1. Separate exact-path investigation: predict the six Status branch products and their sum before running the complete local program. Expected end total is3/4; the remaining1/4 belongs to today paths. For sun/rain, end total is5/8, with3/8 on more branches. Greedy sun then more is a two-step prefix, not a specified completed sequence. No live status or weather measurement is simulated.

   Keep the go/wait/ask record for independent work.

Run the complete exact-arithmetic program

```
from fractions import Fraction as F
first = {'ready':F(1,2), 'busy':F(3,10), 'unknown':F(1,5)}
end_given = {'ready':F(4,5), 'busy':F(1,2), 'unknown':F(1)}
assert sum(first.values()) == 1
paths = {}
for token, probability in first.items():
    paths[(token,'end')] = probability * end_given[token]
    paths[(token,'today')] = probability * (1-end_given[token])
print(paths)
print('all paths:',sum(paths.values()))
print('end paths:',sum(p for (token,next_token),p in paths.items() if next_token=='end'))
# Separate sun/rain tree: more branches are unfinished, not missing mass.
```

```
weather = {'sun':F(55,100),'rain':F(45,100)}
weather_end = {'sun':F(4,10),'rain':F(9,10)}
ends = {token:probability*weather_end[token] for token,probability in weather.items()}
print('sun/rain ends:',ends,'total:',sum(ends.values()))
```

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?: _____**

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

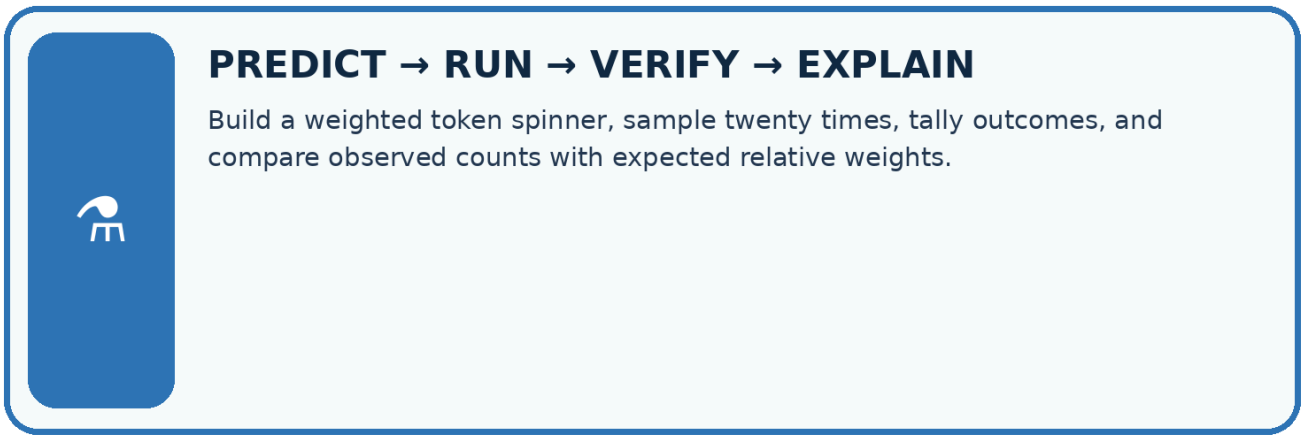
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the nine equal numbered slips with replacement for20draws, recording the actual sequence and tally. Separately run the complete exact conditional-path program. Do not claim that twenty draws must equal expected counts or that generated status labels are

measurements.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the fresh go/wait/ask paths

> Keep greedy first choice, end-path comparison and factual verification separate.

Independent table and scope

First tokens go=0.4, wait=0.35, ask=0.25. End probabilities given them are 0.5,0.8,1 respectively; each branch has a complementary more token. Compute all three end-path probabilities and their sum.

Which first token is greedy, which end path is largest, and does either verify a real status?

Use exactly the stated first-token probabilities and end conditionals. Each more probability is1minus its own end conditional. Compare the specified two-step end paths, not all possible eventual completions. The question supplies no real status observation. Record what those missing observations prevent you from claiming.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Analyse a new table independently

Week 2 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A toy next-token **distribution** assigns cat 0.5, dog 0.3 and bird 0.2. Check the total and name the most likely token. Must it be selected every time when sampling?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result
</details>

Write or draw your reasoning and evidence.

Connect to your project

Explain that Study Buddy responses come from possibilities, so important claims need checks.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A toy next-token **distribution** is sun=.5, rain=.3, snow=.2. Over ten **sample**s, must sun occur exactly five times? Does the highest-probability token establish a true statement about weather?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Explain uncertainty without replacing source checks

> A distribution describes model choices, not a certificate of factual truth.

This week’s concrete contribution

For the Verified Study Buddy, record that each generated token depends on the prefix and decoding rule. Compare factual claims with approved source passages after generation; neither a likely first token nor a high path probability verifies a claim. For the optional representation strand, keep token IDs, embedding rows, position scheme and any aggregation explicit; an equal row sum need not mean equal source order.

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Explain that Study Buddy responses come from possibilities, so important claims need checks.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

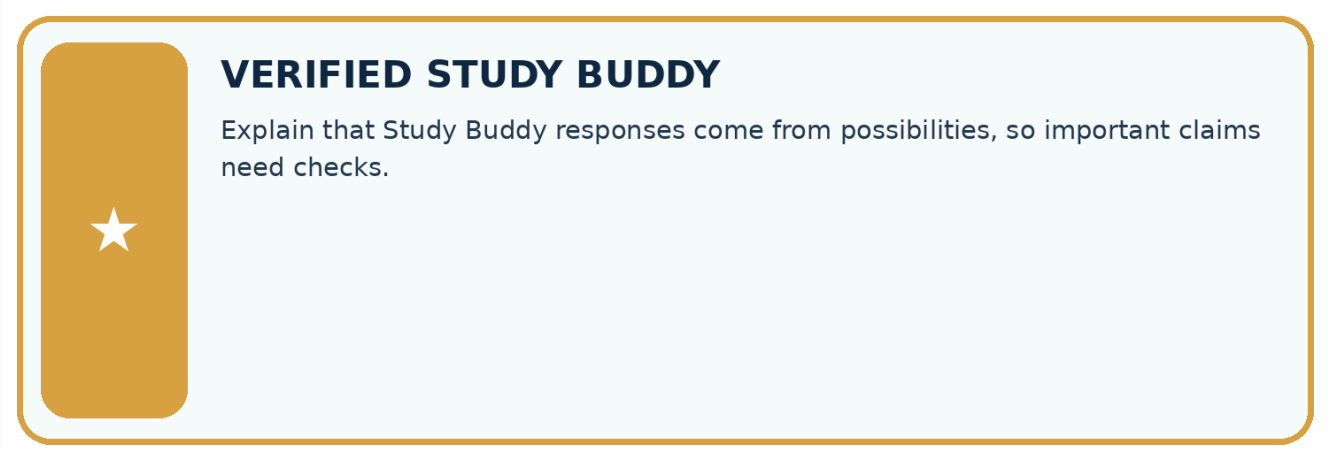
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the prefix, path and evidence boundary

> A path likelihood answers a model question, not a world-state question.

Explain without notes

- Close your notes. Explain this goal in your own words: Check whether a next-token table is a valid **probability distribution**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
distribution	A description of how values or probabilities are spread across possible outcomes.	
likelihood	In this introductory context, how plausible an outcome is under a model; in statistics, likelihood has a more specific parameter-based meaning.	
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.	
continuation	Text that follows an existing sequence.	

Main explanation terms

Word	Meaning
Probability distribution	Nonnegative probabilities over stated alternatives that sum to one

Word	Meaning
Conditional probability	A probability evaluated given specified information
Sequence probability	The product of successive conditional probabilities under an autoregressive model
Probability mass	The probability assigned to an alternative or a set of alternatives

Extension terms

Term	Meaning
Embedding	A vector associated with a token under a specified model
Position vector	A vector carrying information about a sequence position
Representation width	The number of components in each vector
Elementwise addition	Adding corresponding vector components

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

DISTRIBUTION

possible outcomes with relative likelihoods

2

LIKELIHOOD

relative weight assigned to an outcome





Words used in the closer explanation

Terms for Embeddings and position

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

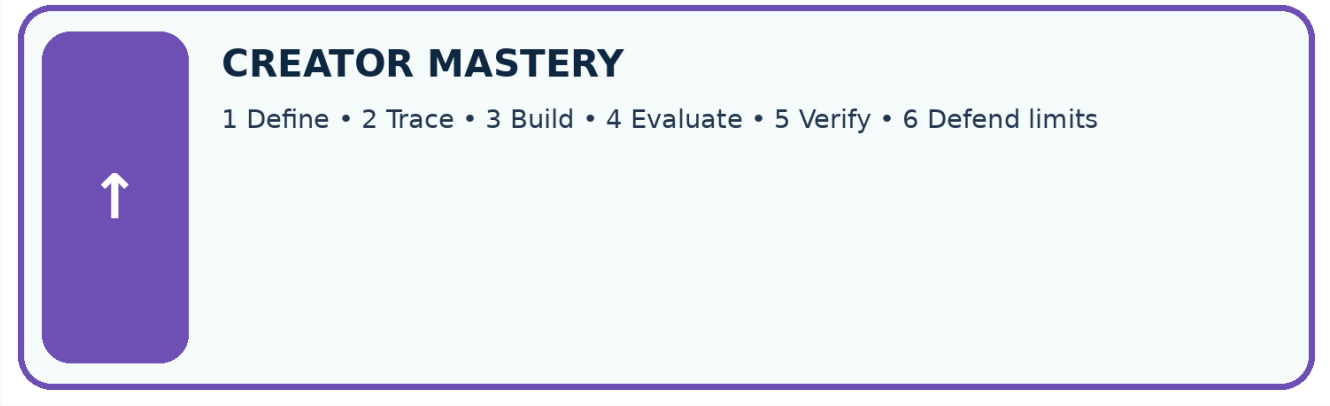
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

REFERENCE · SECTION 12

Reference · Check your explanation

> Use after practice; Audit new conditional paths independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

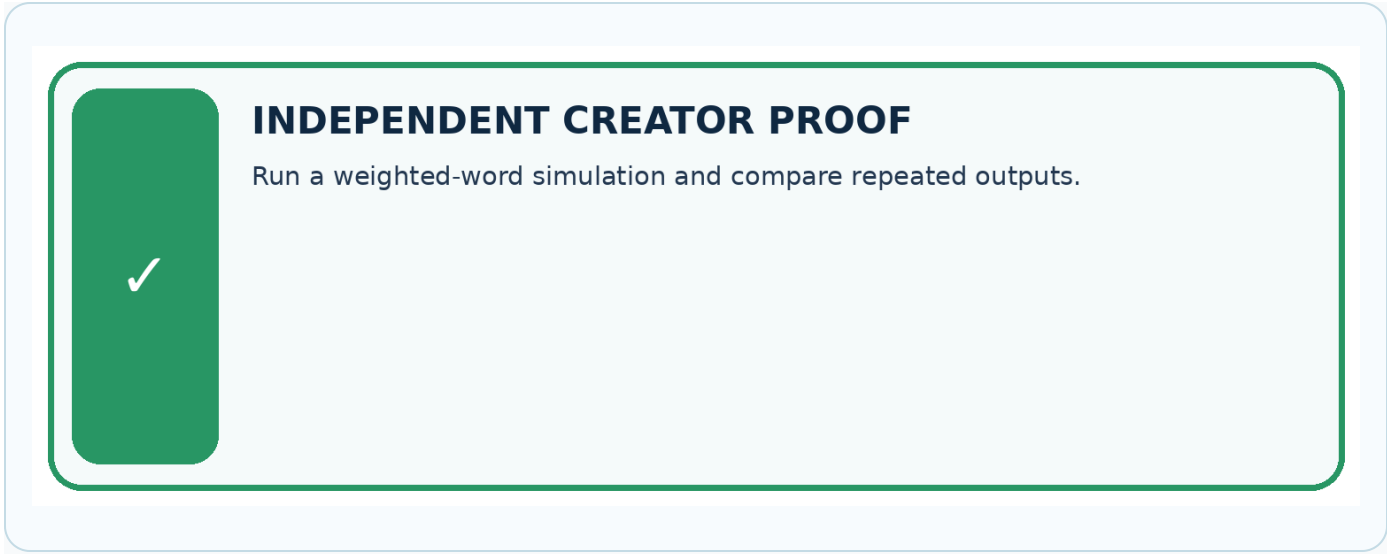
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week's explanation

Use after practice. Audit new conditional paths independently is the selected independent task.



Require the correct prefix, joint products, event sum and limit on factual interpretation.

▼ **Extra practice · The earlier complete case**

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

A toy next-token **distribution** is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Check after trying

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

Evidence or stage	Value or result
rain	50
sun	30
snow	20

Earlier wording, alternatives and picture transcripts

Week 2 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A toy next-token probability distribution

A toy next-token **distribution** is rain .5, sun .3, snow .2.

Check total and interpret the largest value.

Total is 1. Rain is most likely, not guaranteed under sampling. Token probability is not whole-sentence truth.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Check whether a next-token table is a valid probability distribution	Explain the deciding step, show a trace or test, and state one limit.
Calculate a two-step sequence probability using conditional probabilities	Explain the deciding step, show a trace or test, and state one limit.
Explain why token probability does not measure factual truth	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional two-stage token tree illustrates next-token **prediction**.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional two-stage token tree illustrates next-token **prediction**.
- Which labelled step matches this detail? rain→end is the more probable of the two complete end paths.
- 1. Rain branches
- 2. Compare end paths
- 3. Sun branches
- 4. First **distribution**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Compare end paths rain→end is the more probable of the two complete end paths. Conditional probabilities describe what follows a particular prefix.

Multiplying along a path gives its joint probability. The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended.

Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Changed-case question

- Changed-case practice: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. sun→end becomes 0.44, beating unchanged rain→end 0.405.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check changed case**

After your attempt, compare with the supported explanation: sun→end becomes 0.44, beating unchanged rain→end 0.405.

Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Repair and keep evidence

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative **prediction** from generation by inspecting the requested **output**.

Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

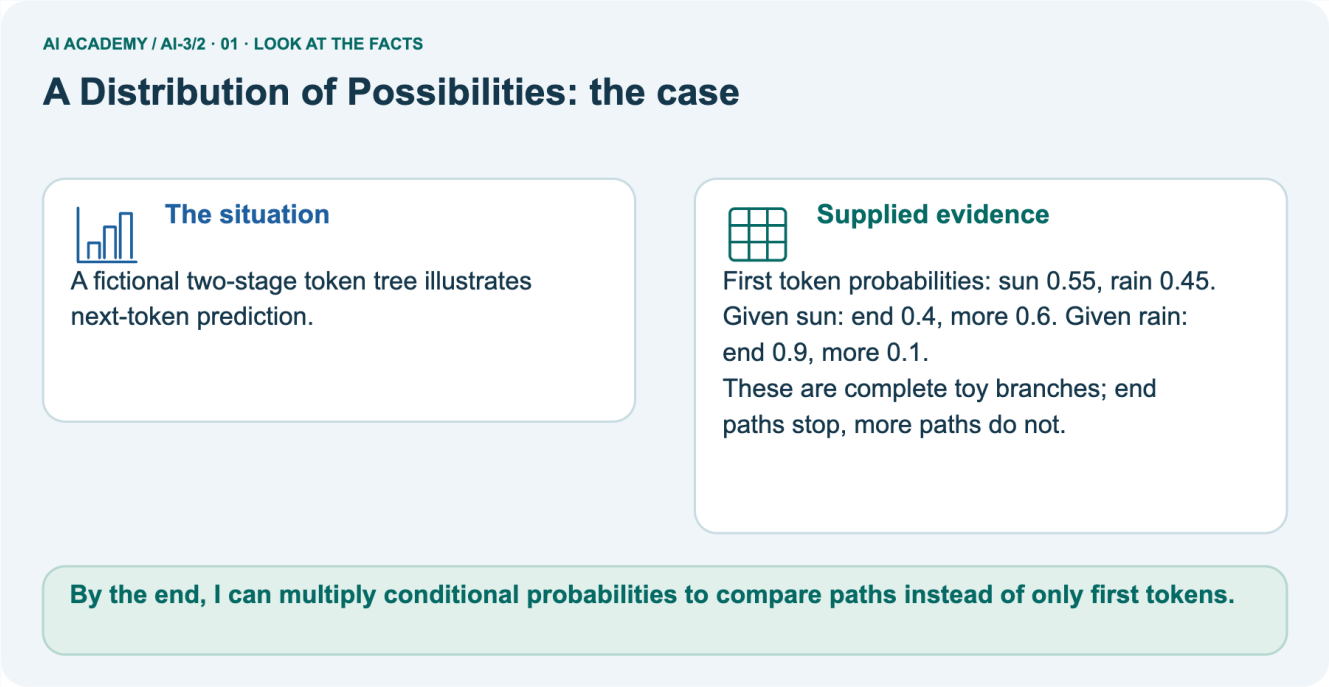
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. A Distribution of Possibilities: the case



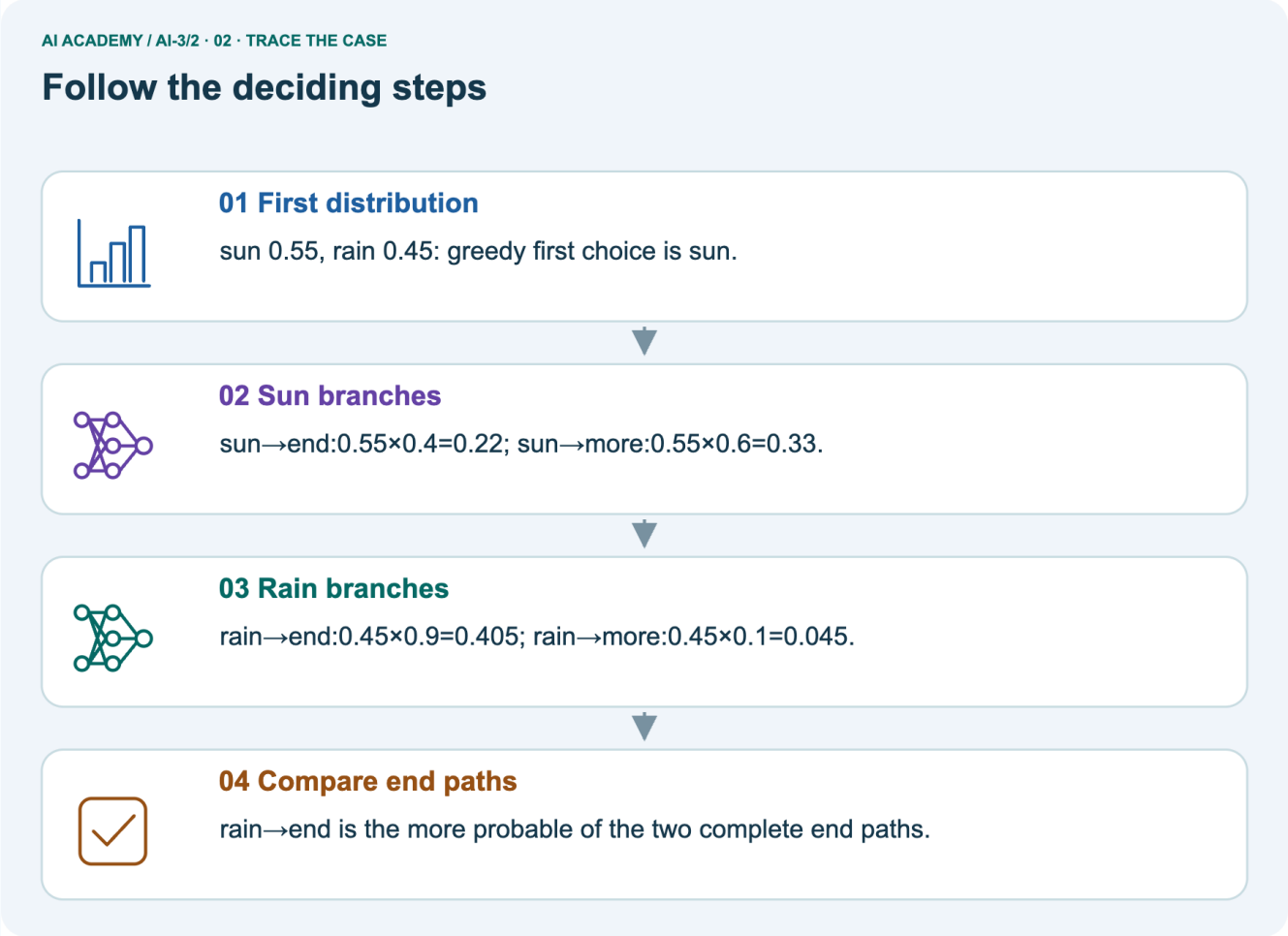
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional two-stage token tree illustrates next-token **prediction**. Supplied facts: First token probabilities: sun 0.55, rain 0.45. Given sun: end 0.4, more 0.6. Given rain: end 0.9, more 0.1.

These are complete toy branches; end paths stop, more paths do not.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

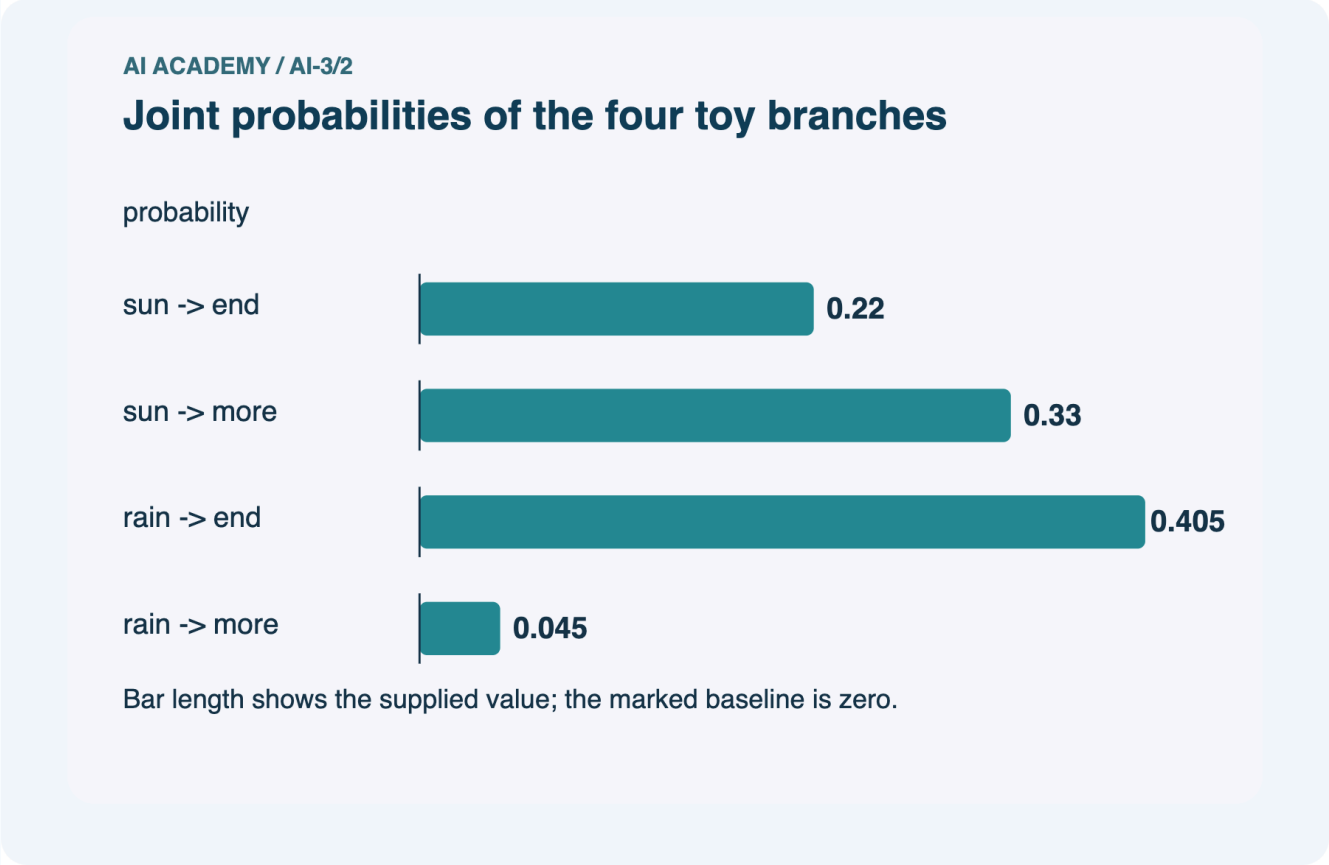


Follow the labelled picture with the matching explanation. Open the vector diagram

First **distribution**: sun 0.55, rain 0.45: greedy first choice is sun. → Sun branches: sun→end:0.55×0.4=0.22; sun→more:0.55×0.6=0.33. → Rain branches: rain→end:0.45×0.9=0.405; rain→more:0.45×0.1=0.045. → Compare end paths: rain→end is the more probable of the two complete end paths.

Point to each numbered stage and explain which recorded **input** it uses.

03. Joint probabilities of the four toy branches

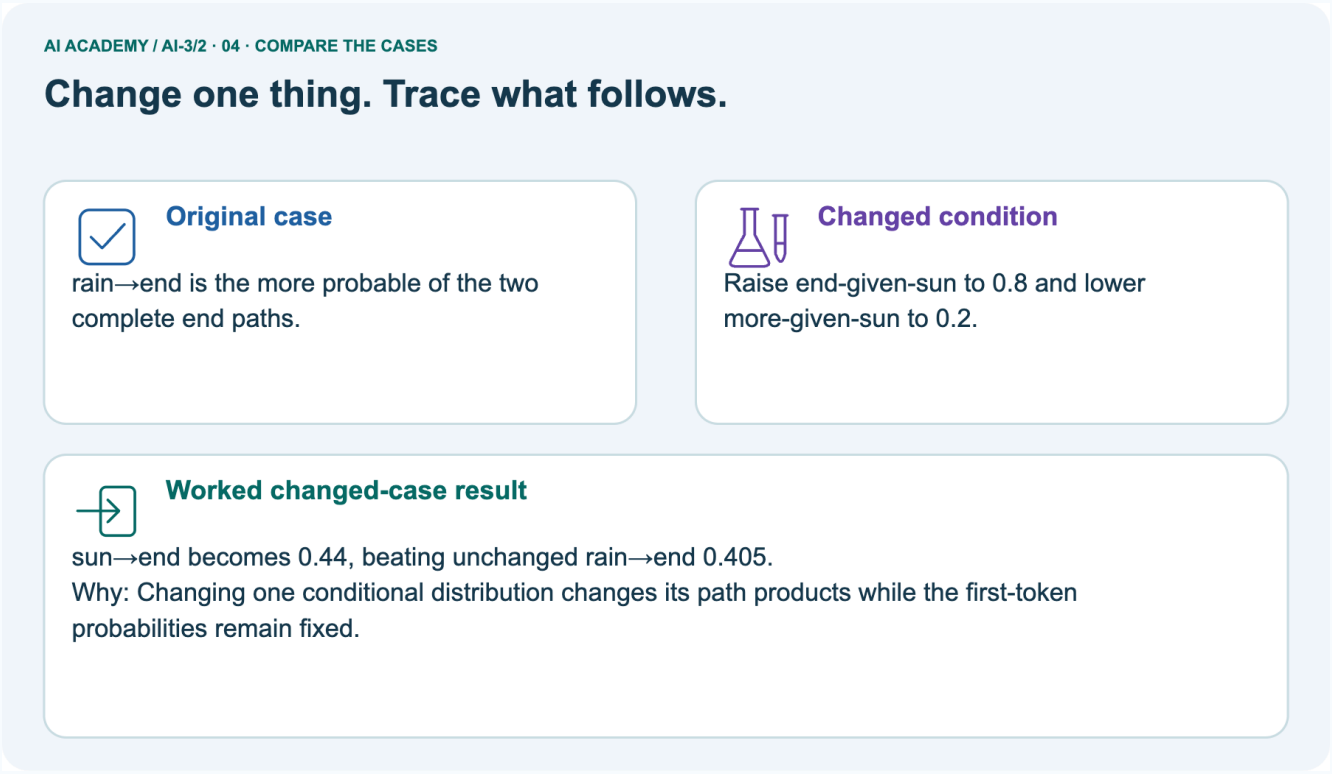


Follow the labelled picture with the matching explanation. Open the vector diagram

Joint probabilities of the four toy branches. Units: probability. sun→end: 0.22. sun→more: 0.33. rain→end: 0.405. rain→more: 0.045.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



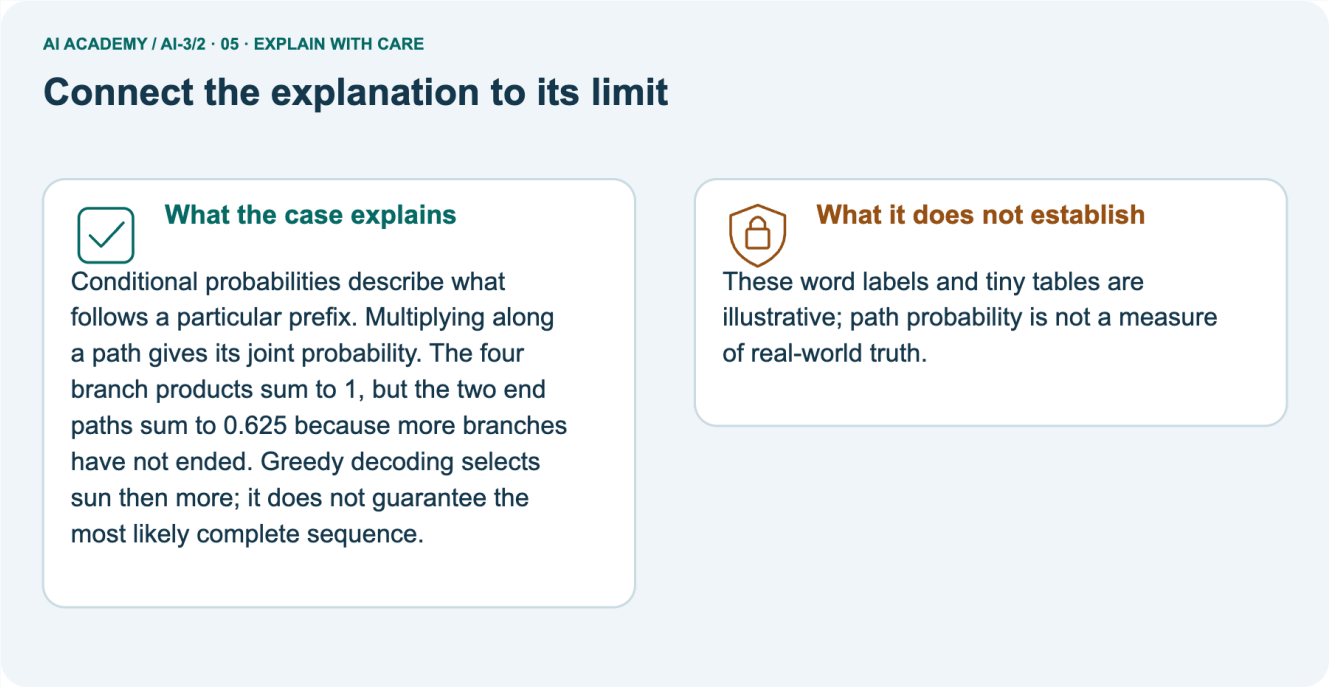
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: rain→end is the more probable of the two complete end paths. Changed condition: Raise end-given-sun to 0.8 and lower more-given-sun to 0.2.

Expected result: sun→end becomes 0.44, beating unchanged rain→end 0.405. Reason: Changing one conditional **distribution** changes its path products while the first-token probabilities remain fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

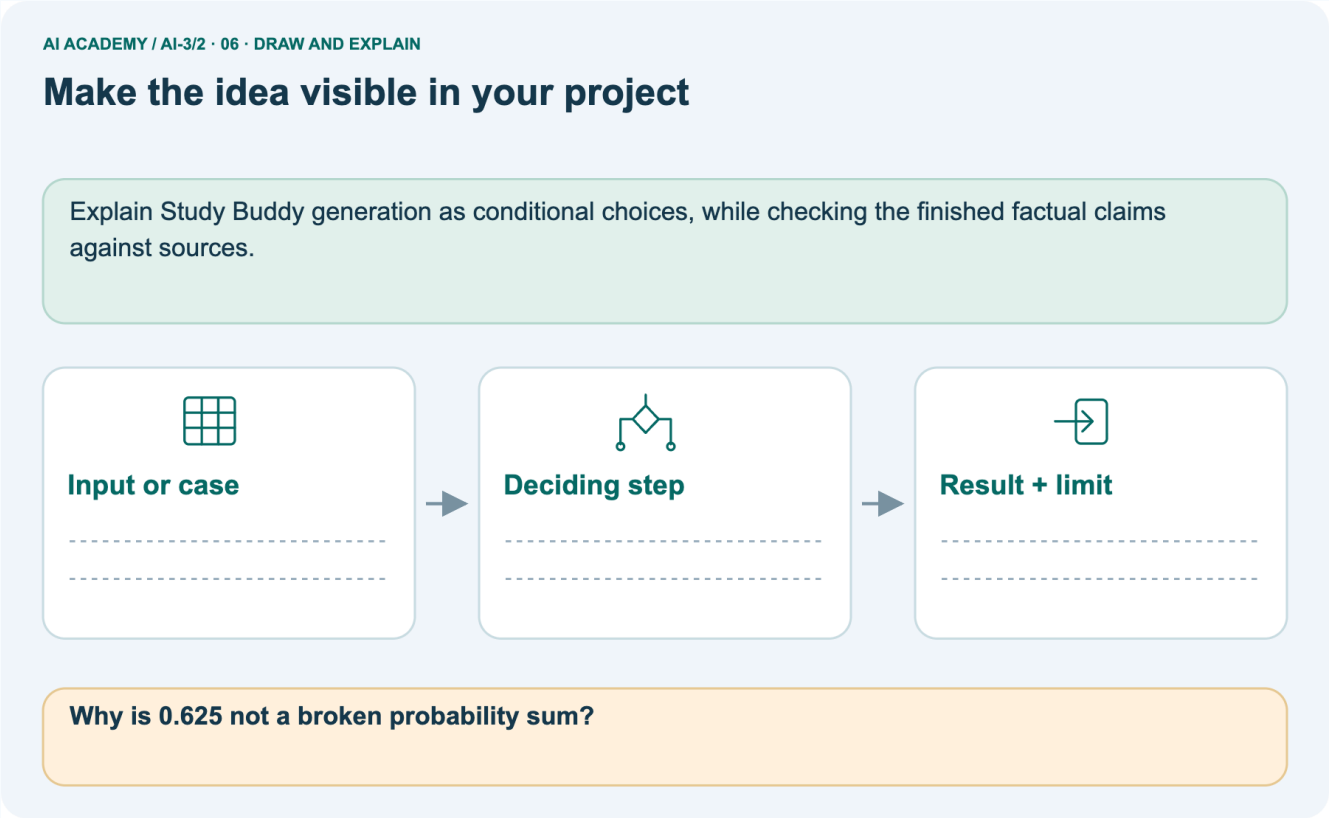
Conditional probabilities describe what follows a particular prefix. Multiplying along a path gives its joint probability.

The four branch products sum to 1, but the two end paths sum to 0.625 because more branches have not ended. Greedy decoding selects sun then more; it does not guarantee the most likely complete sequence.

Limit: These word labels and tiny tables are illustrative; path probability is not a measure of real-world truth.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain Study Buddy generation as conditional choices, while checking the finished factual claims against sources. Draw your **input**, deciding step, result and limit.

Supported question: Why is 0.625 not a broken probability sum?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. A Distribution of Possibilities: the case

02. Follow the deciding steps

03. Joint probabilities of the four toy branches

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.